

Rotating Stand Project Proposal

Semester: Spring 2026

Section: A08 – Group B

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Goal and Summary:

- Create an active motor control to keep the stand within 3 degrees of its upright position
- The stand will need to angle left and right, using a motor to rotate it towards the vertical. The stand will stay upright, even as mass is added to one side of the system to change the balance
- The stand should be able to take impacts and return to an upright position. The force applied to the stand will vary (stand will be pushed by hand)

Inspiration Screenshot:



Self-balancing reaction wheel from [ReM-RC](#).

Controller Type:

- Start with PD controller - main goal is to stabilize the stand. The derivative term helps react to the rotational speed of the stand
- Will implement PID control if there is too much leaning or steady-state error present

Controller Specifications and Requirements:

- Set Point: 0 degrees in vertical/upright position
- Stability duration: Maintain balance for at least 30 seconds
- Rise time: 1 second
- Settling Time: 2 seconds- the device should stabilize and stop major oscillations from a leaning start or small external force
- Steady state error band: +/- 3 degrees from the upright position
- Maximum overshoot: Less than 15%-- the device should not swing past the vertical point excessively

Control Signal/Variable:

- Variable – Angle and angular velocity of the device
 - Goal --> angle close to 0
 - Measured using Adafruit 9-DOF IMU for accelerometer and gyroscope
- Signal – Voltage applied to the DC motor connected to the reaction wheel
 - The voltage signal to the motor applies torque and reaction wheel acceleration
 - Controlled using the Arduino

Bill of Materials: - Jonah

- We have filled out the provided Excel Sheet and sent a screenshot in our MS Teams Chat

A08-B											
Item Description	Cost	Quantity	Extended Cost	Vendor	Link	Section	Link	Notes	Admin		
200 RPM Small 3V DC Motor	\$ -	0	\$ -			A08-B	0	GTAE Item	Y		
Arduino Uno Rev3 SMD			\$ -			A08-B	0	GTAE Item	Y		
Motor Shield for Arduino Uno			\$ -			A08-B	0	GTAE Item	Y		
Adafruit 9 DOF IMU			\$ -			A08-B	0	GTAE Item	Y		
Stock Aluminum			\$ -			A08-B	0	Provided by GT AE Machine Shop	Y		
5V 5A Power Supply			\$ -			A08-B	0	GTAE Item	Y		
12 AWG Solid Core Wire		24	\$ -			A08-B	0	GTAE Item	Y		
Breadboard			\$ -			A08-B	0	GTAE Item	Y		
Multimeter			\$ -			A08-B	0	GTAE Item	Y		
JBWeld		1	\$ -			A08-B	0	GTAE Item	Y		
Total Cost		25	\$ -								

Lab Space Usage

- Loewy Library, ESM G1 (during lab hours) for general meetings
- AMS for 3D printing

AE Shop Usage

- The main reaction wheel and any weights will be manufactured from stock found in the GTAE Machine Shop courtyard, which is free to use
- Jonah is AE Machine Shop trained and able to use the waterjet and mill

Four Mandatory Technical Components:

- **System Dynamics Modeling:** physics-based modeling of the dynamics of applying torque to the reaction wheel and the angular speed of the wheel, and angular speed, position and acceleration of the entire system.
- **Simulation:** the simulations for this model will be done using Simulink
- **Controller Design and Implementation:** PD controller will be designed using Simulink and the controlSystemDesigner app, using the Arduino package to translate to Arduino.
- **Physical Hardware:** Arduino Uno Rev 3 SMD, Adafruit 9-DOF Orientation IMU, 200 RPM Small 3V DC motor, Wheels